Gateway wiring: ESP32 + Ra-02 (8 wires + capacitor)

3.3 V only. The radio goes to 3V3, never VIN or 5V. Power the board only after checking every wire against this page.

1. Connections (this is what to wire)



Capacitor 100-470 µF: **long leg (+) to Ra-02 3.3V**, **stripe leg (-) to Ra-02 GND** (the second GND pin, next to 3.3V).
Antenna: u.FL plug clicked onto the socket, whip screwed on, before power.

2. Where the pins are

ESP32 (USB end at top)

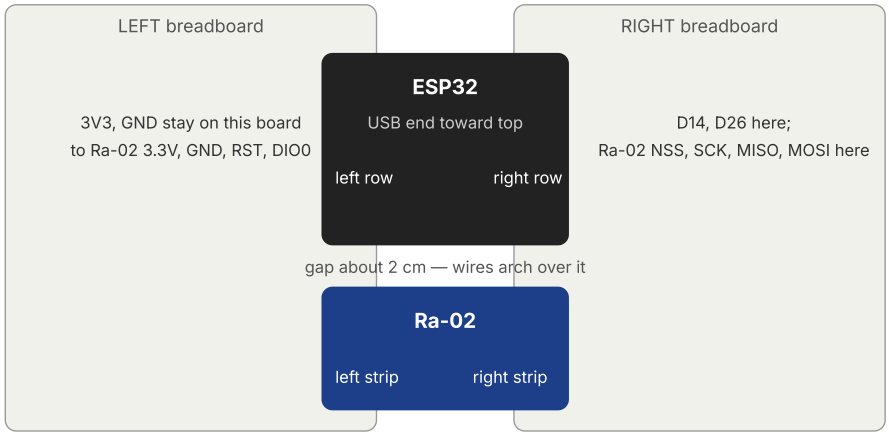
	Left row (left breadboard)		Right row (right breadboard)	
1	**3V3**	USB end	VIN	1
2	**GND**		GND	2
3	D15		D13	3
4	D2		D12	4
5	D4		**D14**	5
6	RX2		D27	6
7	TX2		**D26**	7
8	**D5**		D25	8
9	**D18**		D33	9
10	**D19**		D32	10
11	D21		D35	11
12	RX0		D34	12
13	TX0		VN	13
14	D22		VP	14
15	**D23**		EN	15

Ra-02 (u.FL socket at top-left)

	Left strip (left breadboard)		Right strip (right breadboard)	
1	**GND**	u.FL socket	GND	1
2	GND		**NSS**	2
3	**3.3V**		**MOSI**	3
4	**RST**		**MISO**	4
5	**DIO0**		**SCK**	5
6	DIO1		DIO5	6
7	DIO2		DIO4	7
8	DIO3		GND	8

Coloured cells are used. Grey cells stay unconnected. The right-hand GND pins and VIN are not used.

3. Breadboard layout



4. Check before plugging in USB

- Red wire goes from ESP32 **3V3** to Ra-02 **3.3V**. Black goes from ESP32 **GND** to a Ra-02 **GND**. Do not swap these two.
- Orange, yellow, green, blue go to NSS, SCK, MISO, MOSI. Check MISO and MOSI are not swapped.
- White goes to RST and purple to DIO0, on the *left* strip.
- Nothing is on VIN or 5V.
- Expected result after flashing the gateway firmware: serial monitor prints **LoRa ready**.